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Department of Computer Science & Engineering

Report on Mini Project

Electricity Bill Consumption Prediction

Course Code : CS2003-1

Course Name : Machine Learning

Semester: VI SEM **Section: B**

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CERTIFICATE

This is to certify that Disha Shetty, bearing USN NNM23CS066, of VI Semester B.Tech (Computer Science and Engineering) at NMAM Institute of Technology, Nitte, has successfully completed his Machine Learning mini-project titled "Electricity Bill consumption Prediction" during the period January 2026 – April 2026, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

Name and Sign of Mentor

Signature of HOD

ABSTRACT

Electricity consumption is increasing rapidly due to the growing use of electrical appliances in homes, industries, and commercial buildings. Managing electricity usage efficiently has become important to reduce energy waste and control electricity costs. Predicting electricity consumption in advance can help users understand their future electricity usage and plan their energy consumption more effectively.

In this project, machine learning techniques are used to predict electricity bill consumption based on historical electricity usage data. The dataset is first pre-processed by cleaning the data and selecting relevant features related to electricity consumption. The prepared dataset is then divided into training and testing sets to build prediction models using different machine learning algorithms.

Several machine learning algorithms such as Linear Regression, Decision Tree, and Random Forest are applied to predict electricity consumption. The performance of these models is evaluated using evaluation metrics such as Mean Absolute Error (MAE) .The results show that machine learning models can effectively predict electricity consumption and help users manage their energy usage more efficiently.

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Chapter 1

INTRODUCTION

Electricity is an essential resource used in homes, industries, and commercial areas for operating various electrical appliances and devices. With the increasing use of electronic equipment, electricity consumption has grown rapidly. Monitoring and managing electricity usage has become important to reduce energy waste and control electricity expenses. Predicting electricity consumption can help users understand their future electricity usage and plan their energy usage efficiently

Electricity consumption prediction is an important application of machine learning. By analysing historical electricity usage data, machine learning models can identify patterns and predict future electricity consumption. This helps households and organizations estimate their electricity bills in advance and manage their energy usage more effectively.

In this project, electricity bill consumption prediction is performed using machine learning techniques. The dataset is pre-processed to remove unnecessary data and prepare it for model training. Machine learning algorithms such as Linear Regression, Decision Tree, and Random Forest and KNN are used to build prediction models. The performance of the models is evaluated using appropriate evaluation metrics, and the results show that machine learning can effectively predict electricity consumption.

Chapter 2

PROBLEM STATEMENT

Electricity consumption in households varies depending on several factors such as the number of appliances used, usage patterns, and other energy-related variables. It becomes difficult for users to estimate their electricity consumption and future electricity bills accurately. Manual analysis of electricity usage data is time-consuming and inefficient when large datasets are involved. Therefore, there is a need for an automated system that can analyze household energy data and predict electricity consumption. This project uses machine learning techniques to analyze historical electricity usage data and build predictive models that can estimate electricity consumption more accurately.

OBJECTIVES

1. To analyze electricity consumption data using machine learning techniques.
2. To preprocess and prepare the electricity usage dataset for model training.
3. To apply machine learning algorithms such as Linear Regression, Decision Tree, and Random Forest and KNN for electricity consumption prediction.
4. To evaluate and compare the performance of different models using metrics such as Mean Absolute Error (MAE)
5. To identify the best model for predicting electricity bill consumption.

Chapter 3

RESULT

In this project, electricity consumption data was analyzed using machine learning techniques to predict electricity bill usage. The dataset was first preprocessed by cleaning the data and selecting the relevant features required for prediction. After preprocessing, the dataset was divided into training and testing sets to build and evaluate the prediction models.

Different machine learning algorithms such as K-Nearest Neighbors (KNN) and Random Forest were applied to train the models using the dataset. These models analyzed the historical electricity consumption patterns and generated predictions for future electricity usage. The performance of the models was evaluated using appropriate evaluation metrics.

The results showed that the Random Forest model provided better prediction performance compared to the other algorithm used in this project. The prediction results demonstrate that machine learning techniques can effectively analyze electricity consumption data and estimate electricity bill usage. This can help users better understand their electricity consumption and manage their energy usage more efficiently.

Figure 1: Implementation of Electricity Consumption Prediction Model

1. Importing Libraries and Loading Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("C:/Users/Disha/OneDrive/Desktop/billconsumption/Household energy unit data.csv")
df
```

	num_rooms	num_people	housearea	is_ac	is_tv	is_flat	num_children	is_urban	units
0	3	3	742.57	1	1	1	2	0	134.731117
1	1	5	952.99	0	1	0	1	1	125.315527
2	3	1	761.44	1	1	1	0	0	123.047874
3	0	5	861.32	1	1	0	0	0	80.046162
4	1	8	731.61	0	1	0	2	1	128.429094
...	...	...	...	...	...	...	...	...	...
995	3	0	991.31	0	1	0	1	1	128.128283
996	2	5	847.97	0	1	0	0	0	85.209782
997	2	4	660.94	1	0	0	0	1	117.267814
998	1	1	814.75	0	1	1	0	1	127.823576
999	1	6	835.43	0	0	0	0	1	113.424783

1000 rows × 9 columns

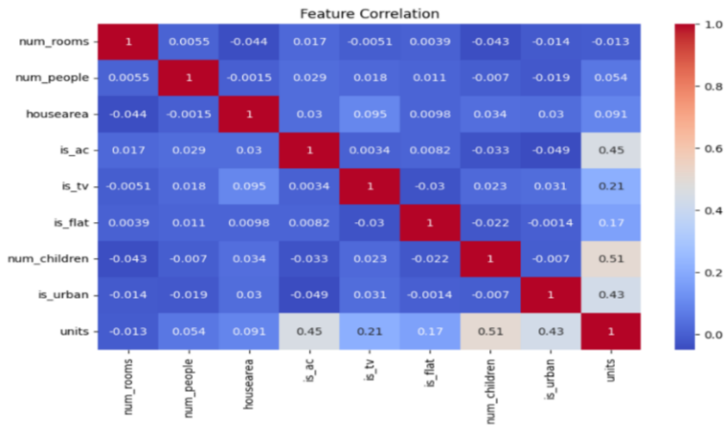
2.Data preprocessing

```
df.isnull().sum()
```

```
num_rooms       0
num_people      0
housearea       0
is_ac           0
is_tv           0
is_flat         0
num_children    0
is_urban        0
units           0
dtype: int64
```

3.Correlation heatmap

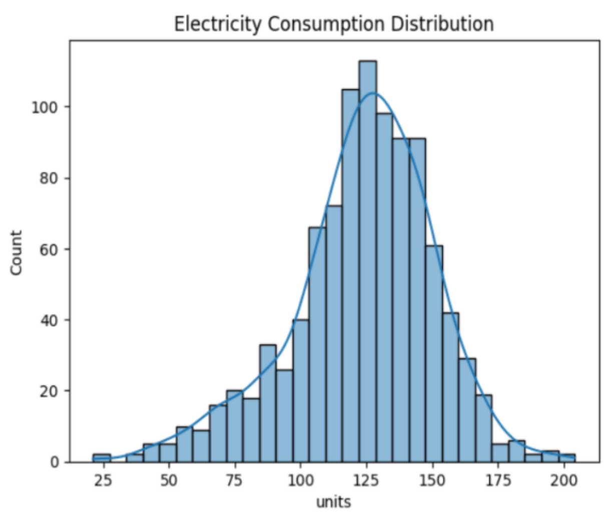
```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Feature Correlation")
plt.show()
```



4.Histplot

[28]:

```
sns.histplot(df["units"], kde=True)
plt.title("Electricity Consumption Distribution")
plt.show()
```



•[29]:

4.Algorithms

```
]: # Split dataset

   from sklearn.model_selection import train_test_split

   X_train, X_test, y_train, y_test = train_test_split(
       X, y, test_size=0.2, random_state=42
   )
```

```
]: from sklearn.preprocessing import StandardScaler

   scaler = StandardScaler()

   X_train = scaler.fit_transform(X_train)
   X_test = scaler.transform(X_test)
```

```
]: from sklearn.neighbors import KNeighborsRegressor
   from sklearn.ensemble import RandomForestRegressor

   # KNN
   knn = KNeighborsRegressor(n_neighbors=5)
   knn.fit(X_train, y_train)
   knn_pred = knn.predict(X_test)

   # Random Forest
   rf = RandomForestRegressor()
   rf.fit(X_train, y_train)
   rf_pred = rf.predict(X_test)
```

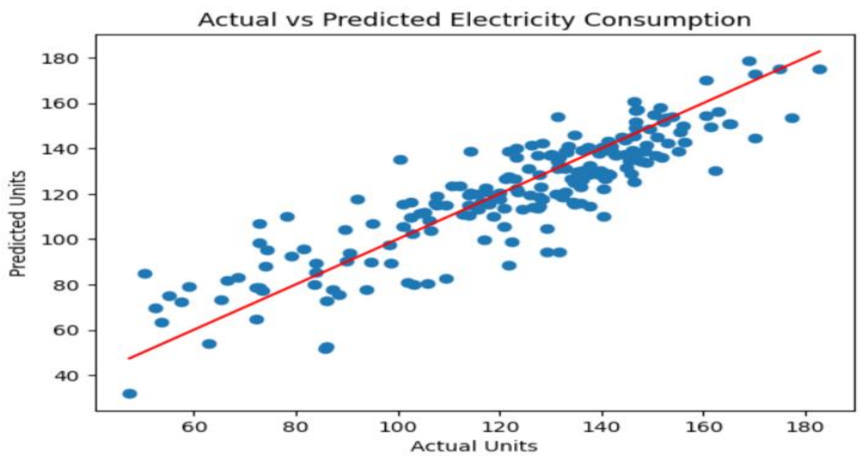
```
from sklearn.tree import DecisionTreeRegressor
from sklearn.linear_model import LinearRegression

# Decision Tree
dt = DecisionTreeRegressor()
dt.fit(X_train, y_train)
pred_dt = dt.predict(X_test)

# Linear Regression
lr = LinearRegression()
lr.fit(X_train, y_train)
pred_lr = lr.predict(X_test)
```

```
[53]:
from sklearn.metrics import mean_absolute_error

print("KNN MAE:", mean_absolute_error(y_test, knn_pred))
print("Random Forest MAE:", mean_absolute_error(y_test, rf_pred))
print("Decision Tree MAE:", mean_absolute_error(y_test, pred_dt))
print("Linear Regression MAE:", mean_absolute_error(y_test, pred_lr))
```

```
KNN MAE: 23.176917706462312
Random Forest MAE: 10.843136200660794
Decision Tree MAE: 13.62125860934673
Linear Regression MAE: 10.82193499701273
```

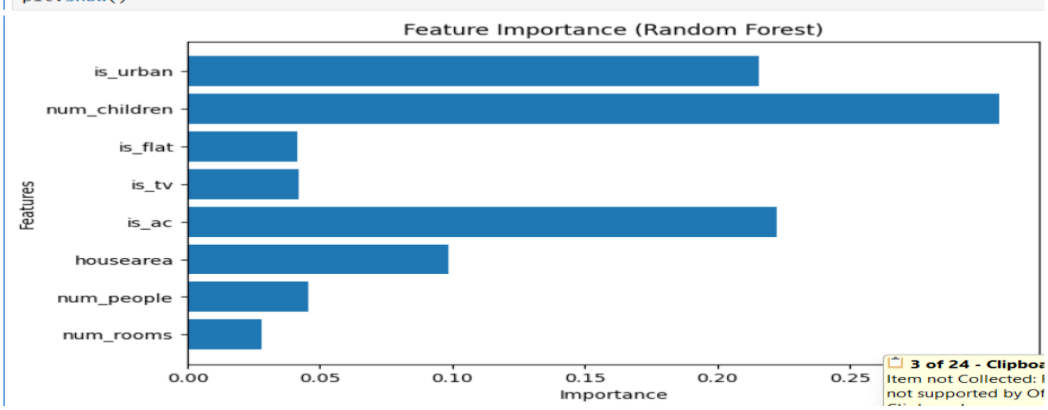
5.Actual vs Predicted Consumption

```
plt.scatter(y_test, rf_pred)
plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()],
         color='red')

plt.xlabel("Actual Units")
plt.ylabel("Predicted Units")
plt.title("Actual vs Predicted Electricity Consumption")
plt.show()
```



5.Features

```
importance = rf.feature_importances_
features = X.columns

plt.figure(figsize=(8,5))
plt.barh(features, importance)

plt.xlabel("Importance")
plt.ylabel("Features")
plt.title("Feature Importance (Random Forest)")
plt.show()
```



6.Final output

```
# Take inputs from user
num_rooms = int(input("Enter number of rooms: "))
num_people = int(input("Enter number of people: "))
housearea = float(input("Enter house area: "))
is_ac = int(input("AC present? (1 = Yes, 0 = No): "))
is_tv = int(input("TV present? (1 = Yes, 0 = No): "))
is_flat = int(input("Is it a flat? (1 = Yes, 0 = No): "))
num_children = int(input("Enter number of children: "))
is_urban = int(input("Urban area? (1 = Yes, 0 = No): "))

# Create dataframe
new_house = pd.DataFrame({
    "num_rooms":[num_rooms],
    "num_people":[num_people],
    "housearea":[housearea],
    "is_ac":[is_ac],
    "is_tv":[is_tv],
    "is_flat":[is_flat],
    "num_children":[num_children],
    "is_urban":[is_urban]
})

new_house_scaled = scaler.transform(new_house)
# Predict using Random Forest
prediction = rf.predict(new_house_scaled)
print("Predicted Electricity Consumption:", round(prediction[0], 2), "Units")
```

```
Enter number of rooms:  2
Enter number of people:  5
Enter house area:  1200
AC present? (1 = Yes, 0 = No):  1
TV present? (1 = Yes, 0 = No):  1
Is it a flat? (1 = Yes, 0 = No):  0
Enter number of children:  2
Urban area? (1 = Yes, 0 = No):  1
Predicted Electricity Consumption: 166.92 Units
```